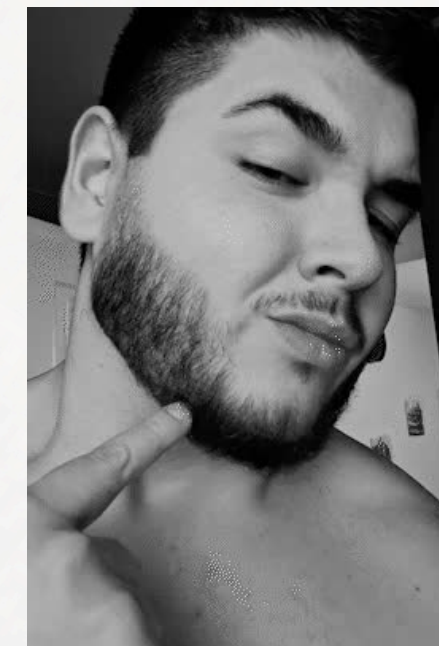




Variable Types and Constants

CMSC 18 | 1S SY2026-27



Zazheska Lorenzo

UP MIN - CSM - DMPCS



Topics for today's session

- Computer Memory
- Variables
- Common Keywords
- Constants



Computer Memory

- **Byte** = the fundamental unit of computer data storage.
- as your program runs, bytes of information is kept in the computer's RAM.
- RAM is **volatile**, which means data is erased and replaced as often as needed.



Variables

- named data storage location in your computer's memory.
- assigns a random range of memory addresses (depending on data type) with the variable name so when you reference the value of variable, it knows where to look.



— Variable Naming Rules

- The name can contain letters, digits, and the underscore character (_).
- The first character of the name must be a letter. Using the underscore is technically legal, but highly unconventional/looked down upon.
- Capitalization matters. Thus, the names “age”, “Age”, and “AGE” are treated as different variables.
- **C keywords** cannot be used as variable names.

Keywords

- words that are pre-built within the language that are assigned to do something.

**Some common
keywords include:**

auto	double	int	struct
break	else	long	switch
case	enum	register	typedef
char	extern	return	union
const	float	short	unsigned
continue	for	signed	void
default	goto	sizeof	volatile
do	if	static	while

Distinguish legal variable names

Which variable names are **legal**, **unconventional**, and **illegal**?

Percent	x3h5o_412ks	case
	Double	
annual_profit		_PROPERTY-Tax
	1990_winter_olympics	

Distinguish legal variable names

Which variable names are legal, unconventional, and illegal?

Percent

x3h5o_412ks

case

Double

annual_profit

_PROPERTY-Tax

1990_winter_olympics

Numeric Variable Types

- different numerical types for variables are necessary because of memory allocation reasons.
- different numerical types are also handled differently in certain mathematical operations.
- Two main categories:
 - Integer Variables
 - Floating-point Variables



Numeric Variable Types

C's numeric data types.

Variable Type	Keyword	Bytes Required	Range
Character	char	1	-128 to 127
Integer	int	2	-32768 to 32767
Short integer	short	2	-32768 to 32767
Long integer	long	4	-2,147,483,648 to 2,147,438,647
Unsigned character	unsigned char	1	0 to 255
Unsigned integer	unsigned int	2	0 to 65535
Unsigned short integer	unsigned short	2	0 to 65535
Unsigned long integer	unsigned long	4	0 to 4,294,967,295
Single-precision floating-point	float	4	1.2E-38 to 3.4E38 ¹
Double-precision floating-point	double	8	2.2E-308 to 1.8E308 ²

¹Approximate range; precision = 7 digits.

²Approximate range; precision = 19 digits.

Variable Declaration

- before you can use a variable, it must exist first. Therefore, declaration.
- variable declaration can be done with the following piece of code:

typename varname[, varname2, ...];

- where **typename** defines the variable's data type, **varname** is the variable's name, and **declaration of additional variables is possible and optional**.

Variable Declaration (cont.)

For Example:

```
int thisIsAnInteger;
```

One Integer Variable

```
int number1, number2;
```

Two Integer Variables

```
float percent, net, gross;
```

Three Float-point Variables

Variable Initialization

- providing value to the variable declared.
- can be done during declaration.

int count = 2;

double percent = 0.01, taxrate = 28.5;

char symbol = 'X', letter;

Numeric Variable Initialization

- be careful not to assign a value to a numeric data type that is outside its intended value range (refer to slide 10).

int weight = 100000;

unsigned int val = -10;

unsigned char val = 258;

Constants

- Unlike a variable, the value stored in a constant CANNOT be changed during program execution.
- C has 2 types of constants:
 - Literal Constants
 - Symbolic Constants



— Literal Constants

- a value that is typed directly into the source code wherever it is needed.

int count = 2;

float num = 21.420;

values 2 and 21.420 are the literal constants.

Literal Constants

- a number constant written without a decimal point is assumed by the compiler to be an integer.

72 100 45

- a number constant with a decimal point is assumed by the compiler to be a double-precision value.

100. .21 31.45

Symbolic Constants

- a constant that is represented by a name.

```
#define PI 3.14159;  
  
int radius = PI + 10 * 3;
```

PI is the symbolic constant

```
const double PI = 3.14159;  
  
int radius = PI + 10 * 3;
```



The End

THANK YOU FOR LISTENING

Zazheska Lorenzo

UP MIN - CSM - DMPCS